

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.

```
import csv
a=[]
with open('/content/enjoysport.csv') as csfile:
    reader = csv.reader(csfile)
    for row in reader:
        a.append(row)
        print(row)
num_attributes=len (a[0])-1
print(num_attributes)
print("The most specific hypothesis:",["0"]*num_attributes)
print("The most generic hypothesis:",["?"]*num_attributes)
hypothesis=a[1][:-1]
print(hypothesis)
print("\n Find S: Finding a maximally specific hypothesis")
for i in range (len(a)):
    if a[i][num_attributes]=="Yes":
        for j in range(num_attributes):
            if a[i][j]!=hypothesis[j]:
                hypothesis[j]='?'
        print("The training example no:",i+1,"the hypothesis is :",hypothesis)
print("\n The maximally specific hypothesis for training set is")
print(hypothesis)
```